HybridGuard: Cross-Layer Consistency as Integrity Evidence for Android Web Environments

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Abstract

Web services use browser-visible environment reports to recognize clients and inform security decisions, yet reports can be modified without corresponding device changes. In Android hybrid Web applications, Android Native, WebView Host, and App Web provide complementary observations of the same environment. Their semantic relationships offer potential integrity evidence: an individually plausible Web report may be incompatible with its system or container context.

HybridGuard associates observations from the same hybrid Web execution and checks their semantic compatibility, accounting for unavailable evidence and legitimate cross-API differences. We collect these observations and use controlled fingerprint manipulations and separate fixed-check comparisons to examine the effectiveness and limits of cross-layer references.

Across 14 controlled manipulation configurations, HybridGuard's App detector detects 105/126 effective modifications (83.3%), with no alarms on 252 controlled normal records. A Native memory reference detects 18/18 modifications versus 6/18 for a global Web-only threshold, both with 0/48 normal alarms, showing the additional information supplied by device-specific context. A Native timezone reference and an absolute Web condition each detect 6/6 modifications, with 0/30 and 6/30 normal alarms, respectively; those six alarms arise from legitimate system timezone changes. In a separate WebView Host geometry study, a Host reference and a fixed Web-height threshold each detect 6/6 modifications, with 0/66 and 6/66 normal alarms, respectively; the threshold flags legitimate layout changes. These findings support cross-layer consistency as useful integrity evidence when an observable, semantically related reference remains informative beyond the manipulated report. Neither Native nor Host is a root of trust, and consistency neither proves authenticity nor guarantees detection of coordinated cross-surface manipulation.

CCS Concepts

• **Security and privacy → Web application security**; *Intrusion/anomaly detection and malware mitigation.*

Keywords

Android WebView, environment report integrity, cross-layer consistency, device fingerprint manipulation, semantic references

1 Introduction

Web security decisions often depend on client reports about the execution environment. Browser-visible properties can help recognize returning clients or augment authentication [1, 7], but a plausible value can still be fabricated. Targeted browser spoofing shows that an adversary can imitate fingerprint attributes through browser modifications [8]. Assessing report integrity therefore requires evidence beyond the plausibility of individual values.

Android hybrid applications offer complementary observations of the same execution. JavaScript runs inside a WebView whose host and underlying Android environment expose partially overlapping information [2]. A Web memory report may fit a global range yet exceed a device-specific envelope. A legitimate timezone change may move system and Web offsets together. Resizing the container may change webpage geometry without any report manipulation. These examples motivate interpreting Web reports against the context in which they were collected.

Consistency checking is established in fingerprinting research [6, 10, 11]. We ask a more specific question: **when do Native and Host references help distinguish a modified Web report from legitimate environmental variation, and what alarm and availability costs follow?** These references have their own timing, availability, and semantic limits. Their value depends on whether the relevant relationship can be assessed for the current observation.

HybridGuard associates Native, WebView Host, and App Web observations with an execution and evaluates semantic checks under explicit applicability conditions. It constructs compact detectors subject to constraints on normal alarms and definite-output coverage. The Integrated App detector combines within-view conditions with Native–Web memory and timezone relations. We evaluate Host geometry separately because the original App model-construction records lack the paired Host–Web snapshots this relation requires. An optional linked Browser, running in a separate execution context, adds an App–Browser timezone comparison.

In a controlled benchmark of 126 effective modifications and 252 normal record positions, Integrated App detects 105 modifications and a separately refitted Web-only detector detects 96; neither raises a controlled-normal alarm. A refitted no-memory alternative also detects 96, with all nine differences confined to the CPU-plus-memory recipe. Historical normal records reveal costs that this controlled benchmark misses: relaxing the normal-alarm constraint adds three detections but raises alarms on every historical evaluation record. The Browser extension similarly adds targeted detections while increasing unknown outcomes.

We make three contributions:

Conditional evidence semantics. An observation and decision model associates complementary Android measurements with an execution and distinguishes predicate violations, definite non-violations, unavailable evidence, and execution or association failures.

Mechanism evidence. Exploratory fixed-check comparisons examine how device-specific memory bounds, timezone co-change, and Host container context distinguish the tested report edits from legitimate variation. Native memory and timezone are integrated into the App detector; Host geometry remains a standalone study.

WWW '27 submission draft,
2027.

Table 1: Comparison of observation surfaces and integrity-assessment scope in related studies.

Study	Research focus	Observation surfaces				Assessment scope		
		Native	Host	App Web	Browser	Cross-layer checks	Benign change evaluation	Unknown outcome handling
FP-Scanner [10]	Spoofing-induced inconsistencies	×	×	×	✓	×	×	×
Picasso [3]	Canvas-based device classification	×	×	×	✓	×	×	×
FP-Inconsistent [11]	Evasive-bot fingerprint inconsistencies	×	×	×	✓	×	×	×
BrowserFM [6]	Fingerprint dependency modeling	×	×	×	✓	×	×	×
WVProfiler [9]	WebView fingerprint exposure	✓	×	✓	×	×	×	×
Tracking Without Borders [15]	Mobile–Web tracking links	✓	×	✓	×	×	×	×
Cross-Boundary Mobile Tracking [4]	Java-to-JavaScript data flows	✓	×	✓	×	×	×	×
HybridGuard (ours)	**Cross-layer report integrity**	✓	✓[a]	✓	✓[b]	✓	✓	✓

Notes. ✓: explicitly examined within the study's scope; ×: outside its stated focus. Native covers Android app/system-side observations, Host covers WebView container measurements, and App Web covers embedded-Web observations. Cross-layer checks assess Web-report integrity against associated Native/Host references. Benign-change evaluation measures false alarms under legitimate environmental or configuration changes. Unknown-outcome handling distinguishes insufficient or inapplicable evidence from a definite no-alert result. [a]Host geometry is evaluated separately from Integrated App. [b]Browser is an optional linked extension.

Alarm and availability analysis. Refitted detector alternatives and a linked-Browser extension quantify modification coverage alongside normal alarms and definite outputs. Together, these experiments characterize contextual evidence for report integrity within the studied settings.

2 Background, Prior Work, and Scope

2.1 Fingerprint Consistency and Contextual References

Fingerprinting research examines relationships as well as individual attributes. FP-Scanner detects countermeasure-induced inconsistencies, including discrepancies across JavaScript and HTTP reports [10]. Picasso compares canvas challenge responses against device-class expectations [3]. FP-Inconsistent studies spatial and temporal inconsistencies in evasive bot traffic [11]. BrowserFM models dependencies among fingerprints, system characteristics, and browser configurations [6]. Our study builds on this use of consistency as evidence.

WebView research examines both sides of the application–Web boundary. WVProfiler studies fingerprint exposure through Android WebView instrumentation [9]; Tracking Without Borders examines tracking across Java/Kotlin, JavaScript, and runtime behavior [15]; Cross-Boundary Mobile Tracking studies Java-to-JavaScript information diffusion [4]. Our focus is a current Web report's compatibility with associated Android device and container measurements, including relation applicability, missing evidence, and behavior under edits and legitimate changes. Table 1 maps these research scopes.

2.2 Observation Surfaces and Conditional Compatibility

Let E denote the execution state and container context. An observation $x = (N(E), H(E), W(E))$ associates Native system and hardware measurements, WebView Host context, and App Web reports. The App schema has 177 catalogued fields: 84 Native, 26 Host, and 67 App Web. An optional Browser contributes 67 fields from a separate Web execution, yielding a 244-field paired schema when validly associated. Selected detectors use subsets of these fields. Association does not imply atomic sampling, and no surface is trusted ground truth.

A semantic relation C_j constrains observations within a stated applicability domain. A local edit may replace $W(E)$ with $W'(E)$ while a reference stays unchanged; a legitimate change may shift related observations together. Compatibility accounts for differences in API units, precision, and abstractions. The notation E describes the execution context; checks use the collected measurements.

A violation of an applicable relation provides **conditional report-integrity evidence**: the observed values conflict with the declared constraint. The evidence identifies a relational inconsistency, while attribution to an altered operand and the underlying true values remain unresolved. In evaluation, a trigger on a known effective modification counts as a detection.

2.3 Adversary and Claim Boundary

We use controlled browser overrides or emulation to edit resource, browser, locale, display, and graphics reports exposed to the Web application while a compared reference may remain unchanged. An operation counts as effective only when its effect is observed at the target surface. At inference, the detector evaluates each current record using its associated observations and the checks selected offline.

The intended setting is a cooperating Android application that collects Native and Host measurements alongside its embedded webpage. An ordinary remote website needs such integration to obtain these observations. Native and Host measurements are themselves mutable; being outside a local Web override does not make them administratively or statistically independent.

An ALERT indicates a violation of a selected relation, whereas NO ALERT indicates definite non-violation across all selected checks. Both describe the evaluated relations under available evidence, leaving attacker intent and authenticity unresolved.

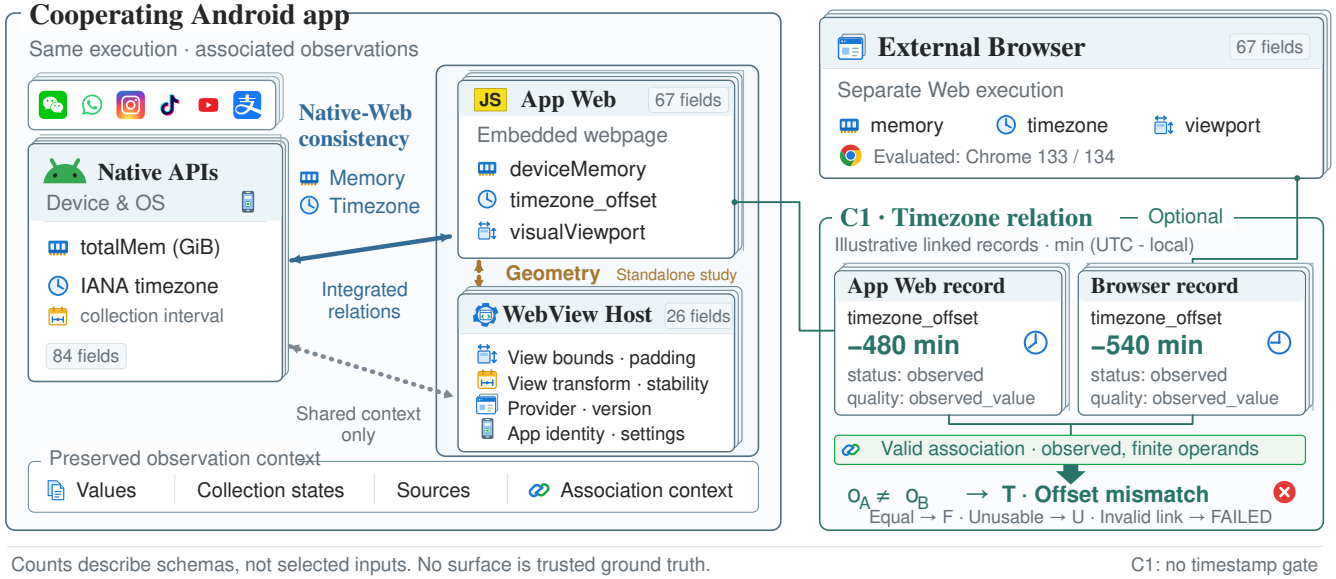


Figure 1: HybridGuard observation surfaces and evaluation roles. Solid blue denotes Native–App Web memory and timezone checks incorporated into Integrated App. Dashed ochre marks Host–Web geometry evaluated separately from Integrated App candidate selection, while dotted Native–Host indicates shared context. A validly linked external Browser enables the optional C1 timezone comparison with App Web. Field counts indicate catalogued schema attributes; detectors select subsets for evaluation.

Local or coordinated edits may preserve these relations. We study controlled Web-report modifications under the stated collector and backend assumptions; collector or backend compromise, adaptive evasion, tracking, and device re-identification fall beyond this evaluation.

3 HybridGuard Design

3.1 Collection and Association

HybridGuard preserves field values, collection states, and association context so that each comparison can be traced to the execution it describes. Source records and receipts support these binding checks. If external Browser collection is unavailable, the collector retains the App record without synthesizing Browser values. Association links observations through recorded context and provenance, supporting comparisons across separately sampled, software-reported measurements.

Collection context also determines applicability. For Host geometry, snapshots immediately before and after the Web snapshot establish Host stability within one collection. These are separate from the pre-intervention and post-restoration records used to evaluate the experiment. The detector uses only the current observation. Figure 2 separates this evaluation path from offline model construction.

3.2 Evidence States and Decisions

Each condition specifies its operands, source requirements, interpretation, applicability domain, and trigger. Its output is $r_j(x) \in \{\mathrm{T}, \mathrm{F}, \mathrm{U}, \mathrm{FAILED}\}$: T is an applicable, assessable trigger; F is a definite non-trigger; U indicates missing, insufficient, or inapplicable evidence; and FAILED records a relevant execution or association failure. An empty selection returns EMPTY_MODEL.

For a nonempty selection, required-input failures take precedence. Otherwise, any T produces ALERT, all F produces NO ALERT, and every remaining combination produces UNKNOWN. We record these three detector outputs as T, F, and U, respectively. A T can therefore coexist with another condition's U but cannot override FAILED. Missing unselected fields do not fail the detector. These outputs characterize the available evidence: F records a definite non-trigger across selected checks, while U preserves unresolved evidence; neither establishes authenticity.

3.3 Semantic References

Native memory envelope. For observed, finite, positive memory measurements in GiB from the same bound App record, with $m_{\text{native}} \geq 0.25$, the condition is

$$r_{\text{mem}}(x) = T \iff m_{\text{web}} > 2^{\lceil \log_2 m_{\text{native}} \rceil}.$$

Here, m_{native} is kernel-visible total memory reported by Android `ActivityManager.MemoryInfo.totalMem`, converted to GiB; m_{web} is the Web `deviceMemory` report. The historical Device Memory API draft describes approximate reporting with implementation-defined bounds [14]. Our power-of-two envelope is an engineering hypothesis that accommodates coarse reporting and permits lower Web exposure. It detects upward violations;

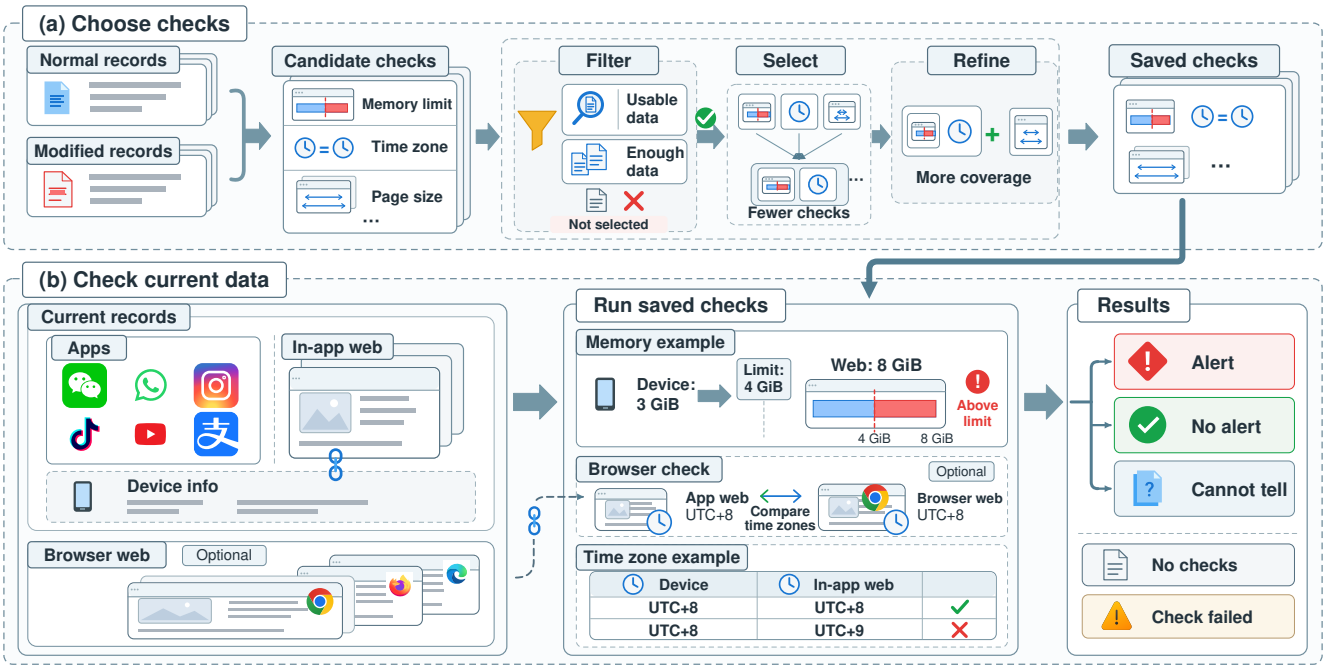


Figure 2: Constrained selection and retention freeze checks for current-record evaluation. App-only and App-plus-Browser are the evaluated configurations; Host geometry remains standalone. Outputs follow the first matching case in Section 3.2.

within-envelope and coordinated modifications can remain compatible. Unusable operands yield U, and binding failures yield FAILED. The fixed Web-only comparator tests $m_{\text{web}} > 8$ GiB. For example, a Web report of 8 GiB exceeds a Native-derived 4 GiB envelope but not the global threshold.

Native timezone reference. The Native IANA zone and collection interval determine the effective UTC offset under the fixed timezone database, 2026c. Let $o_{\text{native}}(t)$ denote local-minus-UTC minutes at the applicable collection time and o_{web} denote Web getTimezoneOffset. The trigger is $o_{\text{web}} \neq -o_{\text{native}}(t)$, accounting for the Web API's UTC-minus-local convention [5]. Supported fixed-offset identifiers can be checked without a date when consistent with the observed Native standard offset. Different zone names with equal effective offsets are compatible. Missing reliable intervals for time-dependent zones, ambiguous transitions, or unusable observations yield U; database or association failures follow the failure policy. The evaluator treats zero and -1 as valid observed offset values.

Host geometry reference (standalone). The Web interface reports visual-viewport dimensions in CSS pixels and a separate scale factor [12]. For each axis, P is the current visual-viewport CSS dimension multiplied by the current device-pixel ratio and `visualViewport.scale`; H_{view} is the measured Host View dimension minus the padding at both ends of that axis. Either width or height triggers when

$$P > H_{\text{view}} + \epsilon(P, H_{\text{view}}).$$

Here, ϵ is a pre-specified engineering tolerance; Appendix A.6 gives its exact definition and post hoc sensitivity analysis. Applicability requires snapshots from the same bound observation, Host stability across the snapshots immediately before and after the Web snapshot, finite positive dimensions, device-pixel ratio and viewport scale, an attached visible View, zero padding, and an identity View transform. Host events and before/after states determine stability independently of Host–Web agreement, allowing stable contradictions to trigger. Missing or inapplicable evidence retains U; axis outcomes follow the OR and failure rules of Section 3.2, including execution and binding errors. F records compatibility with this one-sided upper bound, leaving authenticity unresolved. Host geometry is evaluated as a standalone check outside Integrated App and its candidate pool.

3.4 Constrained Construction and Retention

The App pool contains 50 base templates and three fixed relations: Native memory, Native timezone, and same-Web viewport. The same-Web comparator flags either visual-viewport dimension exceeding the corresponding `window.innerWidth` or `window.innerHeight` by more than 1 CSS pixel, using positive, same-probe observations when visual-viewport scale is at least one. Each template defines a condition over available fields. Training-only encoding and threshold expansion instantiate 65/65/56 atoms before admission in configurations 01/02/03. Admission checks candidate polarity and assessability. Historical evaluation records do not supply threshold quantiles. A literal selects an admitted polarity; each clause contains one literal, and the detector ORs its selected clauses using Section 3.2.

For a set S of singleton clauses, the sparse stage uses

$$J(S) = \frac{1}{14}\sum_{k=1}^{14} d_k(S) - 0.005\, C(S).$$

Here k indexes the 14 manipulation recipes. For each recipe, d_k averages whole-detector detections over the three effective modifications in each training environment, then equally across the two training environments. $C(S)$ counts clauses plus literal occurrences, assigning two complexity units to each singleton clause. With the fixed coefficient 0.005, each clause incurs a penalty of 0.01, equivalent to one percentage point of macro-averaged detection rate. The search uses greedy improving additions, the best feasible prefix, and backward pruning. This heuristic allows at most eight clauses, complexity 16, two conditions per registered family, and no opposite polarities of one atom.

Controlled-normal and historical training-normal alarm budgets are 8/168 and 31/630. Candidate assessability measures the fraction of records yielding T or F for that candidate; definite-output coverage measures the same fraction for the whole detector. Historical candidate assessability and final coverage must each reach 90%. Controlled coverage must reach 80% in each pre, modified, and post phase. Fits allow no execution failures and have a 60-second deadline.

Retention starts from each configuration's sparse solution. Semantic coverage counts distinct (semantic group, modified training record) trigger pairs using the training detection weights, with groups assigned from field semantics. Semantic quality sums predefined preference scores for positive literals, encoding observation preferences that guide retention alongside detection and normal-alarm constraints. Additions must preserve training detections, increase semantic coverage, and satisfy the alarm and coverage constraints. Same-group swaps to registered higher-quality alternatives additionally forbid new controlled-normal alarms or lost definite controlled outputs. Retention seeks additional semantic coverage or higher semantic quality without continuing to maximize J. Each rule-based App configuration is evaluated using its saved retention model; stages are not selected separately for individual result cells.

3.5 Optional Linked Browser

Integrated App + C1 adds a timezone comparison between linked App Web and Browser records. C1 requires observed field status, observed-value quality, and finite numeric non-Boolean operands. Unequal offsets under the shared Web sign convention trigger T; missing operands yield U, and invalid association yields FAILED. C1 evaluates the paired Web offsets directly using a timestamp-independent check, whereas the Native timezone relation resolves offsets using the collection interval.

Extension selection considers no addition, C1, language condition C2, and C1+C2 on the frozen App base. Normal-alarm budgets are 31/630, 0/12, and 1/34 for historical training, pilot, and matched normals. Each normal group must retain 90% definite outputs, and historical candidate assessability must reach 90%. C2 alarms on two matched normal preference changes; C1+C2 also yields only 566/630 definite historical training outputs. We therefore retain C1 under these constraints on development-used material. The selected C1 check augments the frozen App rules, with their outputs combined under the same decision semantics.

Table 2: Evaluation materials and their roles. Counts denote cohort-specific record positions rather than independent devices; study groups have distinct denominators and development histories.

Subset	Δ	N	Role / boundary
Controlled			
App	126	252	Disjoint held-out positions[a]
MTC			
Discovery	—	630	Training-side constraints and tree training
Historical development	—	144	Historical evaluation with development exposure
Historically reserved	—	117	Historical evaluation; not a newly blinded cohort
Specialist			
Memory	18	48	Fixed checks; 6 no-effect operations excluded
Timezone	6	30	Includes 6 normal system changes
Geometry	6	66	Includes 6 normal layout changes; not integrated
Paired			
App–Browser	14	46	60 development-used positions: pilot 18 + matched 42

Δ = confirmed effective modifications; N = normal record positions, including the listed legitimate system and layout changes. [a] Controlled App: 14 manipulation recipes × 3 repetitions × 3 environments (126 modified positions); each has a pre-intervention and post-restoration normal record (252 normal positions).

4 Evaluation Protocol

4.1 Questions and Materials

We evaluate controlled App detection (RQ1), the contribution of semantic references (RQ2), normal alarms and evidence availability (RQ3), and the increment from a linked Browser (RQ4). Table 2 defines each study's denominator and role. Each study's reported rates summarize outcomes across its recorded evaluation positions. Because the studies have different development histories and may overlap in source records or analysis ranges, we report them separately.

The controlled App benchmark uses API 29, 30, and 36 emulators with SwiftShader. Fourteen manipulation recipes, three repetitions, and three environments yield 126 effective modifications. Each run contributes a normal pre-intervention record and a confirmed post-restoration record, yielding 252 correlated normal positions. Model configurations 01, 02, and 03 hold out API 29, 30, and 36, respectively. Each model uses 84 modified and 168 normal training positions, then evaluates 42 modified and 84 normal held-out positions. We aggregate these disjoint held-out positions once. Prior development exposure to the recipes and environments limits claims about unseen tools and physical devices.

Table 3: Controlled benchmark positions: detections among nine effective modifications per original recipe; parentheses show nonzero differences from Integrated App. Only identical complete six-method state-count vectors are grouped; recipe counts retain weights. Ablations are separately refitted.

Modification group	Recipes	Integrated App	No MTC cap	Web-only	No memory	No timezone	App tree
CPU + memory	1	✓ 9	✓ 9	× 0 (−9)	× 0 (−9)	✓ 9	× 0 (−9)
Screen metrics	1	◐ 6	✓ 9 (+3)	◐ 6	◐ 6	◐ 6	× 0 (−6)
Timezone; plugins/MIME	2	✓ 9	✓ 9	✓ 9	✓ 9	✓ 9	× 0 (−9)
Shared detections†	8	✓ 9	✓ 9	✓ 9	✓ 9	✓ 9	✓ 9
Webdriver; WebGL IDs	2	× 0	× 0	× 0	× 0	× 0	× 0
Total detections	14	**105/126**	**108/126**	**96/126**	**96/126**	**105/126**	**72/126**
Detection (%)	—	83.3	85.7	76.2	76.2	83.3	57.1
Missed	—	21	18	30	30	21	54

Tested detections: ✓ all nine; ◐ $0 < d < 9$; × none. Symbols do not rank overall quality. †Playwright, Puppeteer, CDP multi-field, and Stealth presets; platform-only, UA-only, UA-plus-platform, and language-list edits. All methods: 0/252 controlled-normal alarms and zero U, FAILED, and EMPTY_MODEL on these held-out positions. Saved results: RETENTION for rule models, FITTED for the tree. Three extra no-cap screen detections carry a separate normal-record cost (Table 5).

Table 4: Fixed specialist comparisons. Normal denominators include six system changes (Timezone) and six layout changes (Geometry). U and FAILED are zero. Six no-effect memory operations are excluded from 18 effective modifications and 48 normals.

Fixed check	Detected / modified	Normal alarms
Memory		
Web memory > 8 GiB	◐ 6/18	0/48
Native memory envelope	✓ 18/18	0/48
Timezone		
Absolute Web offset	✓ 6/6	△ 6/30
Native timezone reference	✓ 6/6	0/30
Geometry		
Fixed Web height	✓ 6/6	△ 6/66
Same-Web viewport relation	× 0/6	0/66
Host upper bound (standalone)	✓ 6/6	0/66

Detection on tested modifications: ✓ all; ◐ some; × none. △ nonzero normal alarms.

The mobile test-cloud (MTC) collection contains 891 records organized by manufacturer, model, and Android release, with representatives recurring across collection positions. The 630-record discovery group supplies training constraints and tree training; the historical cohorts of 144 and 117 records have prior development exposure and are reported separately. Here, "normal" refers to records without the target manipulation, while debugging and automation may remain part of the collection context.

4.2 Compared Methods

Integrated App combines within-view and cross-view checks. Four alternatives each rebuild their sparse and retention stages: no MTC alarm cap, App Web information only, no memory, and no timezone. Removing the cap relaxes only the alarm budget; MTC assessability and coverage requirements remain. These are comparisons between refitted detectors. Web-only changes both the available information and the relations that can be constructed.

The App tree uses the same fold-specific admitted candidates, encoded as T/F/U indicators. It uses Gini splitting, depth three, and a minimum leaf size of two. Half the training weight goes to modifications, one quarter to controlled normals, and one quarter to MTC normals; modification weight is balanced across the 14 recipes. Together, these settings define a fixed-capacity decision-tree baseline over the admitted candidates. We evaluate saved fitted tree outputs and saved retention outputs for rule models. Specialist checks and Integrated App + C1 are reported separately.

4.3 Measures and Reconciliation

A T decision counts as a detection on an effective modification and as a normal alarm on a normal record. We tabulate F (definite no alert), U (unresolved evidence), FAILED, and EMPTY_MODEL separately, retaining U positions in the corresponding denominators. Model comparisons preserve each study's original record counts and distinguish aggregate totals from record-level predictions. Counts and proportions summarize the observed evaluation positions.

We reconcile saved output-state partitions and recipe-level counts with the reported totals and experiment identifiers. These presentation checks assess the consistency of archived results and their tabulation (Appendix A).

5 Results

5.1 RQ1: Detection and Component Comparisons

Integrated App detects 105/126 controlled modifications with 0/252 controlled-normal alarms (Table 3). The refitted Web-only and no-memory alternatives each detect 96/126; all nine differences occur in the CPU-plus-memory recipe. No-timezone retains 105/126 because an absolute-offset condition substitutes on this benchmark. Removing the MTC cap adds three screen detections, whose normal-data cost appears in Section 5.3. The fixed App tree detects 72/126.

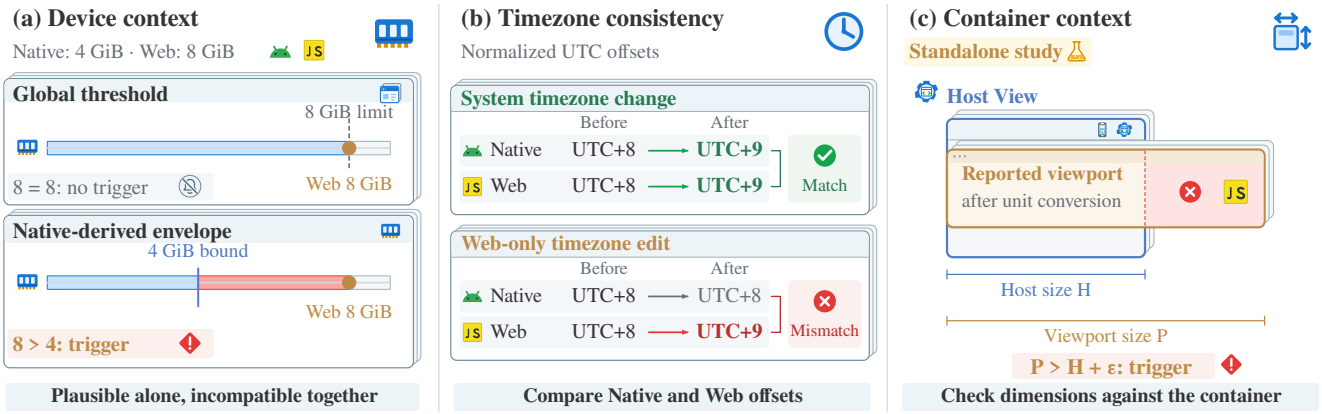


Figure 3: Illustrative mechanisms on applicable observations. (a) A Web report of 8 GiB stays within the fixed 8 GiB comparator but exceeds the envelope derived from Native memory of 4 GiB. (b) Each Native and Web row shows normalized UTC offsets before and after an illustrative change. A system timezone change moves both together, whereas a Web-only edit changes only Web. The right-hand bracket compares the two After offsets; a historical Before input is not required. (c) A converted viewport dimension P exceeding the Host bound plus tolerance triggers the standalone geometry check.

Table 5: Historical-normal decision counts by cohort. Entries report T (alarm), F (definite no alert), and U (unknown). Configurations 01 and 02 share identical per-model counts and one display row. FAILED and EMPTY_MODEL are zero throughout.

Config.	Integrated App	No cap
Historical development ($N = 144$)		
01 / 02	1 / 135 / 8	144 / 0 / 0
03	0 / 136 / 8	144 / 0 / 0
Historically reserved ($N = 117$)		
01 / 02	1 / 110 / 6	117 / 0 / 0
03	0 / 111 / 6	117 / 0 / 0

Bars show the same T / F / U proportions: dark gray / light gray / outlined white. A fully dark bar means every normal record triggered an alarm.

All six methods miss 9/9 webdriver-only and 9/9 WebGL-identifier modifications. Historical raw WebGL unavailability and lost webdriver-false semantics limit the admission of their corresponding candidates. These misses expose gaps in the available feature representation and candidate-admission process for the tested edits. Appendix A describes the recipe grouping, and the supplemental tables retain all 84 recipe-by-method cells.

5.2 RQ2: What References Contribute

The specialist studies evaluate fixed checks separately from the fitted detectors to distinguish three mechanisms (Figure 3; Table 4). Their comparators use Web memory above 8 GiB, Web timezone offset above −480 minutes, and Web height above 710 CSS pixels. The fixed thresholds provide study-specific reference points for comparing the three semantic mechanisms. Appendix A.5 describes their recorded freezing and the scope of these comparisons.

Device-specific memory context. The nine CPU-plus-memory positions separating Integrated App from the refitted alternatives also trigger the fixed 8 GiB check. That main-benchmark difference alone does not establish the need for a device-specific reference. The memory specialist supplies the distinction: the Native envelope detects 18/18 modifications, versus 6/18 for the global threshold, with 0/48 normal alarms for both. Its twelve additional detections are upward edits to 4 and 8 GiB; both checks detect six edits to 16 GiB. Neither triggers on the six no-effect operations targeting 2 GiB. These twelve specialist detections are separate from the nine App-model differences and concern the tested upward edits.

Legitimate timezone co-change. Both checks detect 6/6 modifications. The absolute-offset check also alarms on six normal system changes; the Native reference avoids those alarms. Thus, unchanged main-benchmark detection after removing Native timezone hides a difference in handling legitimate changes. Modification totals alone do not expose this benefit.

Host container context. The fixed Web-height check detects 6/6 modifications but alarms on 6/66 normals. The same-Web viewport relation avoids normal alarms yet misses all six modifications. The Host bound achieves 6/6 detection and 0/66 alarms. This standalone result does not describe Integrated App: configurations 01/02 retain the old height check and alarm on the six normal layout changes, while configuration 03 misses the six modifications. Integrating the Host specialist requires a separate model-construction evaluation.

5.3 RQ3: Normal Alarms and Unavailable Evidence

Removing the historical-normal alarm cap raises controlled detection from 105 to 108 but alarms on every record in both historical groups (Table 5). Both alternatives have 0/252 controlled-normal

Table 6: Paired App–Browser outcomes with development-use history. App = Integrated App; T counts detected modifications or normal alarms. U and FAILED are zero.

Record family	N	App: T	App + C1: T
App timezone	2	✓ 2	✓ 2
App language	2	× 0	× 0
Browser timezone	5	× 0	✓ 5
Browser language	5	× 0	× 0
Normal alarms	46	0	0

Modification rows: ✓ all N detected; × none detected.

alarms, so the controlled benchmark alone conceals this cost. Saved uncapped models include broad conditions such as hardware concurrency greater than one and language-list length greater than one. The controlled-normal budget alone does not adequately constrain these checks on MTC records.

On the separate 630 training normals, Integrated App configurations 01/02/03 produce 6/6/0 alarms and 51 unknowns each; uncapped models alarm on all 630. We keep these training outcomes separate from evaluation. Each Integrated App configuration yields 14/261 unknowns across the two historical evaluation groups. Removing memory reduces that count to 1/261 but also loses resource detections. A tree can return a definite label when some candidate states are unknown, because its traversed path need not use those candidates.

5.4 RQ4: Increment from a Linked Browser

On the 60 paired App–Browser positions used during extension development (14 effective modifications and 46 normals), adding C1 raises detection from 2/14 to 7/14 (Table 6). All five additional detections arise from Browser timezone edits; App timezone detection remains 2/2, while the tested App and Browser language edits remain undetected. Both configurations yield 0/46 normal alarms. These aggregate counts hold for each of the three frozen App models and are presented once for the shared 60-position set.

C1 also raises historical unknowns from 8/144 to 11/144 in development, while leaving 6/117 in the historically reserved group. The combined count increases from 14/261 (5.4%) to 17/261 (6.5%) per model, with unchanged alarms. Asynchronous legitimate timezone changes remain an untested operating condition for C1's timestamp-independent offset comparison. The paired language tests cover App- and Browser-side report edits; the App benchmark separately evaluates a Web language-list modification recipe. C1 provides a targeted increment for the tested timezone edits, accompanied by lower definite-output coverage.

6 Discussion and Limitations

6.1 Conditional Evidence and Evasion

Native and Host observations share the execution environment, so their value as references depends on which observations an adversary can alter independently. Preserving a selected constraint may require control of only one report: a memory edit that remains within the Native-derived envelope can already evade that check. The experiments characterize responses to the tested edits; resistance to adaptive or coordinated evasion and its implications for attacker effort require further evaluation.

6.2 Generality and Development Exposure

The controlled attack evidence covers 14 manipulation recipes across three emulator environments. Repeated runs capture correlated observations within these settings, distinct from independent physical-device samples. Historical normals widen the observed conditions, while prior development exposure in both historical cohorts makes the reserved cohort a retrospective evaluation. Ablations compare separately refitted alternatives. The shallow, fixed-capacity decision tree provides a bounded learning comparator, while Table 1 maps related studies by research scope; performance conclusions are specific to the tested methods and settings.

Development diagnostics show further limits. A relation-augmented tree fits the paired material well but does not retain that benefit across both batch-level development checks. Resource extensions improve some detections but fail the admission constraints. Appendix A and the supplemental tables report these outcomes separately from the retained detector results.

6.3 Availability, Drift, and Integration

Evidence availability shapes both candidate admission and current-record decisions. The missed webdriver and WebGL modifications reveal gaps in the selected representation, while C1 illustrates how an additional reference can reduce definite-output coverage. Changes in APIs, WebView providers, timing, locale conventions, or container behavior may also affect relation applicability. Longitudinal robustness and transfer of the attack results to physical devices require further evaluation. Deployment policies must handle U as unresolved evidence and FAILED as an execution or association failure, treating both as non-definitive outcomes.

The benchmark results include U outcomes, with zero occurrences of FAILED or EMPTY_MODEL. Deployment-oriented evaluation is needed to estimate the frequency of execution failures, rejected associations, and empty selections and to assess response policies for these states.

Host geometry remains a standalone mechanism study. Its current domain excludes nonzero padding and nonidentity transforms. Integrating it would require evaluating detection, alarms, availability, and complexity across window modes, keyboard interactions, and physical devices. The available cached-evaluation timings exclude collection and communication, so end-to-end runtime cost also remains open.

6.4 Data Collection and Privacy

Environment observations collected for integrity checking may also enable tracking [13, 15]. Controlled modifications in this study use emulators, while historical normal records come from cloud-device collection. The MTC collector requests network access and network-state observation; contacts, messages, photos, location, camera, and microphone lie outside its requested permission scope. Fingerprint attributes can nevertheless remain identifying, motivating careful control of their collection and retention.

HybridGuard assesses client-report integrity. Deployments should minimize retained observations and evaluate policies for acting on uncertain evidence. Authentication success, account-takeover prevention, and fraud reduction are downstream outcomes requiring separate evaluation.

7 Conclusion

HybridGuard uses associated Android observations to assess the compatibility of Web environment reports. On the controlled App benchmark, Integrated App detects 105/126 effective modifications with no alarms on 252 normal record positions. Its separately refitted Web-only and no-memory alternatives each detect 96/126; all nine differences occur in the CPU-plus-memory recipe. Exploratory timezone and Host geometry comparisons show how contextual references can distinguish selected edits from legitimate variation. Historical normals and the optional Browser extension expose the accompanying alarm and availability costs. These results support evaluating report-integrity checks jointly by applicability, modification coverage, normal alarms, and definite outputs, within the limits of the references and predicates tested.

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A Configuration and Reproducibility Details

A.1 Saved Model Identities

App configurations 01, 02, and 03 hold out emulator API 29, 30, and 36, respectively. APP_FULL denotes the retained B_REL_TZ models, called Integrated App in the main text. A_NO_MTC_CAP, A_APP_WEB_ONLY, A_NO_MEMORY_REL, and A_NO_TIMEZONE_REL are refitted alternatives. APP_TREE uses saved FITTED outputs; rule models use saved RETENTION outputs. PAIRED_BASE is the frozen App detector plus C1. These names do not add Host geometry to Integrated App.

The candidate-state tree uses scikit-learn 1.7.2, splitter=best, random_state=20261006, no class-weight override, and a positive-class probability threshold of 0.5. Its admitted pools contain 63, 63, and 54 atoms, each encoded by three state indicators; these are smaller than the pre-admission pools. Fifteen positions in each controlled held-out set and 73/891 MTC records have an unknown candidate. A fitted path need not use that candidate, so binary predictions do not imply complete observation.

A.2 Candidate Admission and Extension Choice

Threshold expansion uses training-only quantiles 0.25, 0.5, and 0.75. Candidate support requires at least three complete true/false triplets and at least two triggered modification triplets in at least one bundle/environment, alongside normal assessability. For Browser extension, feasible choices among none, C1, C2, and C1+C2 are ranked by the equal mean detection ratio across App timezone, App language, Browser timezone, and Browser language. Ties prefer fewer additions and then a fixed set order. This ranking is part of development-side selection.

The fixed timezone comparator triggers above −480 minutes; the old Web-height comparator triggers above 710 CSS pixels. Native timezone uses effective offsets under the fixed implementation database. Geometry follows Section 3.3. C1 requires both status and quality markers, accepts finite numeric offsets including zero and minus one, and applies no timestamp gate.

A.3 Full Tables and Statistical Units

S01–S09 retain the nine aggregate tables, including adverse outcomes. Chinese source field names preserve the input schema; main-text tables use English labels. S02 contains 14 recipes × 6 methods = 84 rows. Its five count-pattern groups have sizes 1, 1, 2, 8, and 2, with nine effective modifications per recipe. Weighted sums reproduce 105, 108, 96, 96, 105, and 72 detections. App language-list edits are distinct from the paired study's language operations.

S07 reports resource-admission comparisons on 26 modified and 718 normal development positions, plus a separate replay of 261 historical normals. S0 is the frozen Integrated App + C1 baseline. M adds a Native-to-Browser memory-envelope check, W tests inequality between App-Web and Browser memory reports, and B tests Browser memory above 8 GiB. S0 detects 13/26; S0+M and S0+W detect 19/26; S0+B detects 16/26. These additions fail coverage and/or normal-alarm admission criteria, so they do not replace the retained App-plus-C1 results on 14 paired modifications.

S08 reports a four-view development diagnostic. The four-view diagnostic uses language and timezone measurements: App only, Browser only, their concatenation, and the concatenation augmented by C1/C2 states. P0 trains on the 630 discovery normals and both paired batches. P1 and P2 withhold the 18-position pilot and 42-position matched batch, respectively, while training on the discovery normals and the other batch. The relation-augmented P0 fit yields 14/14 detections and 0/46 normal alarms; under P2, the relation-augmented view yields 4/8 and 6/34. These settings are not independent replications. S09 times cached_from_adapted execution in batches of 60 over ten rounds. Its microseconds-per-position medians and batch-amortized p95 values exclude acquisition and communication; they are neither per-request latency percentiles nor end-to-end costs.

A.4 Verification Scope

Presentation checks verify state partitions, aggregate/detail agreement, row counts, figure and table references, and file hashes. They establish consistency between saved results and the manuscript, without independently validating intervention labels, source-record identity, or collection bindings. The implementation also contains tests for decision states and binding failures. Those tests are distinct from the benchmark outcomes reported here.

A.5 Implementation and Comparator Provenance

The implementation archive provides the 50 base templates and three relations in `deliverables/app177_core_ablation_v1/CANDIDATES.json`, with their dependencies in `CANDIDATE_DEPENDENCIES.md`. Final B_REL_TZ models are saved under `deliverables/timezone_relation_validation_v1/trials/B_REL_TZ__WEBGL1-LOEO-v1-{01,02,03}__RETENTION/model.json`. All three select Native–Web timezone and memory relations, MIME type count above zero, the first-language relation (`RSR-LANG-FIRST-v1`), and the registered UA/platform condition (`CAT:NW-006`). Configurations 01/02 also select Web inner height above 710 CSS pixels; configuration 03 does not. The resulting models have 6/6/5 clauses and complexity 12/12/10.

The active selector, `hybridguard_agent/research/rule_learning_v2/semantic_selection.py`, ranks feasible retention proposals by higher macro-averaged detection gain, higher semantic-coverage gain, fewer controlled-normal alarms, higher minimum phase coverage, lower complexity, higher semantic quality, and finally a structural key. The six saved B_REL_TZ fits terminate through their heuristic stopping criteria, with recorded elapsed times of 0.19–0.48 seconds. The 60-second deadline is applied to each fit and is not their recorded stopping reason.

Record-level outputs are retained in `deliverables/timezone_relation_validation_v1/predictions.jsonl.gz`, `deliverables/app177_core_ablation_v1/results/{full_predictions,predictions}.jsonl.gz`, and `deliverables/cross_endpoint_constrained_extension_v1/results/predictions.jsonl`. These archives support inspection beyond the aggregate tables; saved outputs and independently reproduced results remain distinct forms of evidence.

Comparator provenance differs across studies. The 8 GiB condition was fixed in the memory study's `SETTINGS.json` before the 72-position collection that yielded 18 effective modifications and 48 normal positions; comparisons on the older controlled and MTC records were retrospective. The legacy 710 CSS-pixel height condition appears in the geometry study's `CONDITIONS_FROZEN.json`, fixed before the formal 72-position batch and after six engineering smoke observations. The −480-minute comparator is the legacy absolute-offset condition from the saved B_REL models; old and new model identities were frozen before the 36-position timezone collection. These are controlled mechanism studies conducted during development. The memory envelope and geometry tolerance remain engineering hypotheses within their stated domains. Appendix A.6 adds a post hoc sensitivity analysis of the Host geometry tolerance; memory-envelope sensitivity remains unexamined.

A.6 Host Geometry Tolerance and Post Hoc Sensitivity

The standalone Host geometry check uses the frozen tolerance

$$\epsilon(P, H_{\text{view}}) = 2 + 4 \cdot 2^{-23} \max(|P|, |H_{\text{view}}|, 1) \quad \text{pixels.}$$

The constant 2 allocates one pixel to each of two layout edges as a study-specific quantization assumption. The factor 2^{-23} is the float32 precision scale, and 4 is a fixed coefficient for the numerical margin. This reference scale sets the budget; the saved evaluator computes the comparison using Python `float` arithmetic. The archived rationale leaves an operation-level error derivation for the actual rendering chain open. Establishing a common error bound across WebView implementations requires further implementation-specific analysis. The specification was frozen during development after six engineering-smoke observations and before the formal v15 batch.

We perform a *post hoc sensitivity analysis* using only the 72 formal v15 observations saved in `deliverables/screen_geometry_observation_v1/predictions.jsonl.gz`. The 66 normal records satisfy `normal_basis.supported`; the six effective interventions satisfy `observable_intervention`. These are repeated observations from three emulator environments. Keeping the saved applicability, U/FAILED states, and subset labels fixed, we use each record's current Web operands and measured Host dimensions. We preserve the implementation's multiplication order: first DPR times `visualViewport.scale`, then the CSS dimension times that product. Each axis uses a strict $>$ comparison, and axis outcomes are combined by the original OR rule. With $M = \max(|P|, |H_{\text{view}}|, 1)$, the outcomes are:

Tolerance (pixels)	Normal alarms	Intervention detections
2	36/66	6/6
$2 + 1 \cdot 2^{-23} M$	0/66	6/6
$2 + 2 \cdot 2^{-23} M$	0/66	6/6
$2 + 4 \cdot 2^{-23} M$ (original)	0/66	6/6
$2 + 8 \cdot 2^{-23} M$	0/66	6/6
3	0/66	6/6

For each record, take the maximum of $P - H_{\text{view}}$ over width and height. Its largest normal value is approximately 2.000038 pixels; effective interventions span 475.381–664.381 pixels. Removing the relative term therefore changes decisions for normal observations near the 2-pixel boundary. Coefficients 1, 2, 4, and 8 yield identical decisions on all 72 records, leaving the choice among these coefficients unresolved by this batch. Assessing parameter optimality and performance against small near-boundary interventions requires additional observations. The fixed 3-pixel row serves as a sensitivity comparator; the original coefficient 4 remains the frozen setting. This retrospective analysis follows the original parameter selection, and independent validation of any revised tolerance requires fresh observations.